DOCUMENT INGESTION AGENT

Evaluation Metrics & Rationale

Mortgage Underwriting System

India-focused • Fully automated • No human review in the decision path

Purpose: document what each Document Ingestion Agent metric measures, why it was selected, and how it supports safe, explainable underwriting evaluation when every case is resolved by the agent without human validation.

1. Evaluation Context

The Document Ingestion Agent is evaluated as a fully automated hybrid system. Its spatial extraction core performs multi-format document intake, layout text parsing, structured Pydantic schema extraction, field provenance tracking, deterministic validation, cross-document reconciliation, and finalization. The deterministic layer remains authoritative for all regulatory and financial rules. No case is routed to a human reviewer: every case ends in an automated outcome, and when the evidence is uncertain the agent takes the most conservative outcome, never PROCEED.

Output	Range / Type	Meaning
document_type	Enum (8 classes)	PAN_CARD, AADHAAR, SALARY_SLIP, FORM_16, ITR, BANK_STMT, PROPERTY, UNKNOWN
field_extractions	Pydantic Models	Structured borrower KYC, income, banking aggregates, and property specifications
provenance	Doc + Page + Snippet	Audit trail establishing the exact source evidence for every extracted value
confidence	0–1	Evidence reliability; 0.85 or above is the condition for PROCEED
gate_triggered	e.g. missing_docs	Safety gate that fired; selects the automated action
automated_disposition (new)	Enum (4 outcomes)	Outcome of the case, with no human step

Automated outcome	State	What happens
PROCEED	Terminal	All checks pass and confidence is 0.85 or above; data goes to the credit agent as extracted.
PROCEED_CONSERVATIVE	Terminal	The case continues on the lower, more cautious verified value; the discrepancy is attached so downstream agents see it.
REQUEST_RESUBMISSION	Pending	The agent sends the borrower an automatic request naming the missing, unreadable or conflicting document. The case is parked on a timer and re-enters the pipeline when files arrive.
CLOSE_UNRESOLVED	Terminal	The timeout expires, the maximum number of attempts is reached, or a hard conflict stays unresolved; the case is closed with the reasons recorded.

Conservativeness order, most to least: CLOSE_UNRESOLVED, REQUEST_RESUBMISSION, PROCEED_CONSERVATIVE, PROCEED. When several triggers fire on one case, the most conservative outcome wins. The same order is used to score false accepts and false rejects (section 7).

Termination and re-entry. A pending case becomes CLOSE_UNRESOLVED when the resubmission timeout expires or the maximum number of attempts is reached; both are configuration values (section 7), so every case reaches a terminal outcome. When files arrive, the whole case is re-evaluated on the union of old and new documents, the new result supersedes the old one, and the audit trail is append-only.

Three metric families are required: correctness metrics against labeled ground truth, calibration metrics for the confidence score, and automation and safety metrics measured from the audit trail.

2. Dataset and Ground Truth

The shipped evaluation dataset contains 40 labeled synthetic cases comprising over 160 individual files. Labels are assigned independently from model predictions according to underwriting design intent, avoiding self-confirmation. The 25 edge cases that human review used to catch now each need an expected automated outcome.

Case class	Cases	Rule for the expected automated outcome
Clean Prime Salaried	15	PROCEED
Name Discrepancy	8	Minor spelling variant: PROCEED after re-match. Genuine conflict: REQUEST_RESUBMISSION.
Income Variance	7	Within the variance limit: PROCEED_CONSERVATIVE. Beyond it: REQUEST_RESUBMISSION.
Missing Docs	6	REQUEST_RESUBMISSION
Low-Quality Scans	4	Recoverable by design: PROCEED. Otherwise: REQUEST_RESUBMISSION.

- Each case receives exactly one expected outcome; the table gives the rule for choosing it. The benchmark scores the first-pass outcome, because no borrower reply is simulated.
- Borderline cases are deliberately included, including minor spelling variants and variable payslip bonuses.
- Safety-gate cases include missing_mandatory_docs, salary_discrepancy, name_contradiction, and unmasked_aadhaar.
- The source does not say which name cases are spelling variants or which income cases exceed the variance limit, so those labels must be set per case. This is a one-time labeling task, not a review step in the live pipeline.
- Add cases with forged or spliced documents; an unattended system has no one to notice them (section 10).
- For production evaluation, supplement the synthetic labels with historical underwritten outcomes.

3. Classification Metrics

Classification metrics evaluate whether the predicted document type matches the labeled ground truth across all 8 classes. None of them depend on a reviewer.

Metric	Definition	Why chosen
Accuracy and balanced accuracy	Correct predictions ÷ total cases; mean of per-class recall.	Package class mixes are unequal. Balanced accuracy stops a payslip-heavy mix hiding weak Form 16 or Property results.
Per-class precision, recall, F1	Precision = TP ÷ (TP + FP); recall = TP ÷ (TP + FN); F1 is their harmonic mean.	Categories have unequal business consequences; BANK_STMT recall is the key safety guardrail.
Macro and weighted averages	Equal weight per class; weight by class support.	Macro gives a class-fair view; weighted reflects the real case mix.
Confusion matrix	Actual versus predicted type across all 8 classes.	Shows which error occurred: Bank read as Salary Slip is far worse than a scan sent to UNKNOWN.
Cost-weighted error	Σ cost[truth][prediction] ÷ total documents.	Errors are asymmetric; BANK read as SALARY has the highest prototype cost because it can book unverified credit.

Class	Precision tells us	Recall tells us
PAN_CARD	How cleanly PANs are identified; limits false KYC.	Whether mandatory tax identity proof is captured.
AADHAAR	How reliably resident Aadhaar cards are identified.	Whether mandatory residential address proof is captured.
SALARY_SLIP	Limits false income proofs; avoids unstructured noise.	Whether all submitted monthly earnings periods are caught.
FORM_16	How justified annual tax certificate extractions are.	Whether statutory annual income proofs are caught.
ITR	How cleanly income tax returns are identified; limits false income proofs.	Whether filed-return income evidence is captured.
BANK_STMT	How accurately multi-page transaction ledgers are parsed.	Whether banking cash flows are caught; the most safety-critical recall.
PROPERTY	How cleanly collateral title deeds are identified.	Whether property collateral evidence avoids omission.
UNKNOWN	How often a file sent to UNKNOWN is truly unclassifiable; every wrong UNKNOWN triggers a borrower request.	Whether unclassifiable files are caught instead of forced into a wrong class.

The source report listed only six classes here; ITR and UNKNOWN are added so all 8 classes are covered.

The evaluation uses the following asymmetric prototype cost matrix (PAN = PAN_CARD, SALARY = SALARY_SLIP, BANK = BANK_STMT):

Truth \ Pred	PAN	AADHAAR	SALARY	FORM_16	BANK	PROPERTY	UNKNOWN
BANK	8	8	**10**	6	0	8	2
PAN	0	4	8	6	8	8	2
AADHAAR	4	0	8	6	8	8	2
SALARY	8	8	0	3	**10**	8	2
PROPERTY	8	8	8	8	8	0	3

`Cost-weighted error = Σ cost[truth][prediction] ÷ total documents.`

Without a reviewer: UNKNOWN used to be cheap because a human picked it up. It now triggers REQUEST_RESUBMISSION, so it stays cheap but costs the borrower a delay. Every other misclassification is uncaught, so the high costs describe real exposure. The matrix has rows only for BANK, PAN, AADHAAR, SALARY and PROPERTY as true classes and no ITR column; add rows for FORM_16, ITR and UNKNOWN and an ITR column before treating cost-weighted error as complete.

Important: the cost matrix is configurable and represents an engineering evaluation choice, not an official regulatory cost schedule.

4. Extraction & Validation Metrics

Extraction metrics evaluate whether granular fields and ledgers are extracted verbatim with full source provenance. A failed check now selects an automated action instead of waiting for a person.

Metric	Definition	Why chosen	Automated action on failure
Character and word error rate (CER, WER)	Levenshtein distance to the reference ÷ reference length.	Separates OCR degradation from downstream parsing rules.	Feeds the low-confidence recovery passes.
Exact match (EM)	1.0 if the normalized value equals the label, else 0.0.	Required for discrete identifiers such as PAN (ABCPS1234F) and IFSC (HDFC0001234).	A failed format check lowers confidence and triggers re-extraction.
Token F1	Token overlap between predicted and labeled value.	Tolerates variants such as “Limited” versus “Ltd” in employer names.	None; used for scoring only.
Salary ledger reconciliation	|Gross – (Net + Deductions)| within ±2.00 (rupees assumed).	Forged payslips often fail simple arithmetic; deterministic checks trap fraud before credit scoring.	Payslip excluded from income; REQUEST_RESUBMISSION.
Provenance coverage	Extracted fields with a valid (Doc, Page, Snippet) ÷ all extracted fields.	Keeps every value traceable for audit and for downstream agents. Target 100%.	A field without provenance is re-extracted or dropped, never passed on.

5. Cross-Document Reconciliation Metrics

Reconciliation metrics evaluate whether the agent cross-examines data across independent documents. Contradiction recall is the most safety-critical metric and matters more now: a contradiction the agent misses goes straight to the credit agent with no one to stop it.

- **Name and income reconciliation accuracy.** Names are matched fuzzily across KYC, salary slips and bank statements; monthly net salary is reconciled against detected bank credits within ±5% tolerance.
- **Contradiction detection recall.** True contradictions detected ÷ total actual contradictions. Target benchmark is 98.0% or higher.
- **Contradiction precision.** Real contradictions ÷ contradictions flagged. A false flag sends a resubmission request to a borrower with a clean file.
- **Contradiction handling accuracy (new).** Detected contradictions where the automated action matches the expected one ÷ all detected contradictions.

The benchmark has at most about 15 contradiction cases (8 name, 7 income variance). A perfect recall score on that few cases is still consistent with a true miss rate of up to about 20% (rule of three: 3 ÷ n), so read 100% on this dataset as a floor, not proof.

6. Calibration Metrics

Calibration asks whether the confidence score means what it claims. A confidence of 0.90 should correspond to roughly a 90% probability of extraction correctness. Confidence of 0.85 or above is the condition for PROCEED and is now the only gate between an extraction error and the downstream agents.

What counts as correct. The source does not define correctness for Brier and ECE. Classification accuracy is 0.9938 and field exact match is 0.9770, yet every populated bin below shows 100% accuracy, so the baseline appears to score correctness at the level of the case outcome. Define it explicitly: a case is correct when its automated outcome equals the expected outcome and every field used downstream is within tolerance of its label.

Brier score = mean squared error between confidence and correctness (1 correct, 0 incorrect). Lower is better.

ECE = Σ (|B_m| ÷ N) × |acc(B_m) − conf(B_m)| across 5 confidence bins. 0 is perfect calibration.

Gate-passing error rate (new): cases that are not correct and have confidence of 0.85 or above ÷ all cases with confidence of 0.85 or above. This is the rate at which errors get through the gate unseen.

Confidence interval	Case count	Mean confidence	Actual accuracy	Calibration gap
0.00 – 0.20	0	0.1000	0.0000	0.0000
0.20 – 0.40	0	0.3000	0.0000	0.0000
0.40 – 0.60	0	0.5000	0.0000	0.0000
0.60 – 0.80	**12**	0.6973	1.0000	0.3027
0.80 – 1.00	**28**	0.9142	1.0000	0.0858

- **Under-confident, not over-confident.** All 12 cases in the 0.60–0.80 bin fall below the 0.85 threshold and were correct; they now enter recovery or a resubmission request, which is avoidable friction.
- **ECE needs a re-run.** Weighting the two populated gaps by case count gives 0.3 × 0.3027 + 0.7 × 0.0858 ≈ 0.151, while the baseline reports 0.1489. At 0.151 the 0.150 target is just missed.
- **Gate failures are invisible.** Every populated bin has 100% accuracy, so the table cannot show whether a high-confidence extraction is ever wrong. Add wrongly extracted cases to the benchmark.
- **Proposed:** tighten the ECE target to 0.050 and the gate-passing error rate to 0; recalibrate confidence (for example isotonic regression or Platt scaling) on a larger held-out set and choose the 0.85 threshold from that data.
- **Proposed:** gate on the lowest confidence among decision-critical fields (name, PAN, net salary, bank credits, property value) as well as the overall score, so one weak field cannot be averaged away.

7. Automated Outcomes & Safety Metrics

These replace the straight-through-processing and HITL escalation metrics. Each safety gate maps to an automated action; no gate suspends a case for human review.

Trigger	Automated action	Resulting outcome
missing_mandatory_docs	Send the borrower the list of missing document types and park the case on a timer; no agent can create a missing document.	REQUEST_RESUBMISSION, then CLOSE_UNRESOLVED on timeout or after the maximum attempts.
salary_discrepancy (net salary vs bank credits outside ±5%)	Re-extract both figures once; if the gap remains, use the lower of payslip net salary and the average of detected salary credits over the same months.	PROCEED_CONSERVATIVE; above the salary variance limit, REQUEST_RESUBMISSION.
name_contradiction	Re-match using deterministic normalization only (initials, name order, transliteration); the similarity threshold is never lowered.	PROCEED if resolved; otherwise REQUEST_RESUBMISSION, then CLOSE_UNRESOLVED.
unmasked_aadhaar	Redact the first 8 digits before any extracted value, provenance snippet or log entry is written, so the full number is never stored.	PROCEED, if no other trigger fired.
Ledger arithmetic mismatch	Treat the payslip as unreliable, exclude it from income and request a replacement.	REQUEST_RESUBMISSION.
Confidence below the threshold (including low-quality scans)	At most two recovery passes: image preprocessing and re-extraction, then a second independent extractor compared field by field; confidence is recomputed after each.	PROCEED if confidence recovers to 0.85; otherwise REQUEST_RESUBMISSION for a rescan.

Configuration parameters. Every threshold the flow depends on is listed here so none is left implicit.

Parameter	Used for	Value
Confidence threshold	Condition for PROCEED	0.85 (source)
Income reconciliation tolerance	Net salary versus bank credits	±5% (source)
Ledger tolerance	Gross versus Net + Deductions	±2.00, rupees assumed (source)
Recovery passes per case	Low-confidence recovery	2 (preprocessing, second extractor)
Salary variance limit	PROCEED_CONSERVATIVE versus REQUEST_RESUBMISSION	To be set; not defined in the source
Resubmission timeout	Pending case to CLOSE_UNRESOLVED	To be set; not defined in the source
Maximum resubmission attempts	Guarantees every case terminates	To be set; not defined in the source
Circuit-breaker limits	Allowed drift from the rolling baseline (section 10)	To be set; not defined in the source

Cases that depend on a value marked “To be set” cannot be scored until it is fixed.

Metric	Definition	Proposed target
automation_rate	Cases that reach an outcome (including REQUEST_RESUBMISSION) with no human touch ÷ all cases.	100%
stuck_case_rate	Cases with no outcome, or pending past the timeout ÷ all cases.	0
disposition_accuracy	Cases whose outcome equals the expected outcome ÷ all cases.	0.95 or higher
false_accept_rate	Cases whose outcome is less conservative than expected ÷ cases whose expected outcome is more conservative than PROCEED.	0 on the benchmark
false_reject_rate	Cases whose outcome is more conservative than expected ÷ cases whose expected outcome is less conservative than CLOSE_UNRESOLVED.	Track the trend
edge_case_containment	Edge cases whose outcome is at least as conservative as expected ÷ edge cases (25 in the benchmark).	25 of 25
conservative_value_error	PROCEED_CONSERVATIVE cases where the value used exceeds the verified value ÷ PROCEED_CONSERVATIVE cases.	0
recovery_rate	Cases reaching PROCEED after recovery ÷ cases entering it.	Track the trend
resubmission_resolution_rate	Resubmission requests answered and resolved before timeout ÷ requests sent.	Track the trend
auto_proceed_rate	Cases ending PROCEED ÷ all cases; replaces stp_rate.	Informational

False accepts and false rejects are scored on the conservativeness order in section 1. A metric whose denominator is empty is reported as n/a, not as 0 or 1. Unchanged and label-free: gate_trigger_counts, mean_confidence, below_threshold_rate and provenance_coverage; below_threshold_rate now measures how often recovery starts. Retired: stp_rate and hitl_escalation_rate. On the benchmark the expected auto_proceed_rate is the share of cases labeled PROCEED: at least 15 of 40 (37.5%), more if recoverable scans and spelling variants are labeled PROCEED. Do not push it up at the expense of false_accept_rate.

8. Current Baseline and Targets

Correctness baselines carry over from the original benchmark. The new metrics have no baseline until the automated flow is built and run, and recovery and recalibration may shift the extraction and calibration numbers, so re-run all of them after the change.

Evaluation Metric	Baseline Value	Target Benchmark	Note
Document Classification Accuracy	0.9938 (99.38%)	0.950 or higher	Unchanged
Classification Balanced Accuracy	0.9956 (99.56%)	0.950 or higher	Unchanged
Asymmetric Cost-Weighted Error	0.0124	0.150 or lower	Cost matrix incomplete
Field Extraction Exact Match (EM)	0.9770 (97.70%)	0.920 or higher	Unchanged
Field Extraction Token F1 Score	0.9878 (98.78%)	0.950 or higher	Unchanged
OCR Character Error Rate (CER)	0.0142 (1.42%)	0.030 or lower	Unchanged
Contradiction Detection Recall (Safety)	**1.0000 (100.00%)**	0.980 or higher	Too few cases to certify
Contradiction Precision	1.0000 (100.00%)	0.900 or higher	Unchanged
Brier Score	0.0346	0.100 or lower	Unchanged
Expected Calibration Error (ECE)	0.1489 reported; about 0.151 from bins	0.150 or lower; proposed 0.050	Re-run
Straight-Through Processing (STP) Rate	0.3750 (37.50%)	None	Retired
HITL Escalation Rate	0.6250 (62.50%)	None	Retired
Automation Rate	Not yet measured	100%	New
Stuck-Case Rate	Not yet measured	0	New
Disposition Accuracy	Not yet measured	0.95 or higher (proposed)	New
Edge-Case Containment	Not yet measured	25 of 25	New
False-Accept Rate	Not yet measured	0	New
Gate-Passing Error Rate	Not yet measured	0	New

These values are baseline results for the shipped benchmark dataset and should be interpreted as an engineering benchmark.

9. Why This Metric Set Was Chosen

- **Safety:** contradiction recall, false-accept rate, confusion matrices, and cost-weighted error emphasize the asymmetric harm of corrupted borrower financials.
- **Class fairness:** balanced accuracy and macro metrics prevent payslip-heavy document distributions from hiding weak classification on rare deeds.
- **Extraction fidelity:** Exact Match, Token F1, and CER test whether granular entities and financial figures are extracted without distortion.
- **Trust in automation:** Brier, ECE, reliability bins, and the gate-passing error rate test whether confidence is reliable enough to be the only gate for PROCEED.
- **Automation health:** automation rate, stuck-case rate, disposition accuracy, and recovery and resubmission rates show whether the agent resolves cases correctly on its own.
- **Auditability:** provenance coverage guarantees full source verification rather than treating the LLM/OCR as an unverifiable authority.

10. Risks and Safeguards

The current benchmark cannot show that an unattended agent is safe; it can only show that nothing has failed yet.

- **Too few cases.** The baseline figures are consistent with 161 documents and a single error that cost 2 (160 ÷ 161 = 0.9938; 2 ÷ 161 = 0.0124), so one document moves accuracy by about 0.6 points. With at most about 15 contradiction cases and no wrongly extracted cases, perfect scores say little about the true error rate. Grow the benchmark with adversarial and wrongly extracted cases and add historical underwritten outcomes.
- **Digital tampering is now required, not future scope.** A forged document that is internally consistent passes every check in this report, and no reviewer would see the splice. Quantify precision and recall on image splicing, font inconsistencies, and PDF metadata modifications.
- **Loosened matching.** An automated re-match can “resolve” a genuine conflict by relaxing the matcher. Limit it to deterministic normalizations and never lower the similarity threshold.
- **Sensitive data.** Unmasked Aadhaar numbers must be redacted before they reach provenance snippets, logs or the audit trail, not only in the final stored copy.
- **Downstream inheritance.** Credit and decision agents now receive data no person has looked at. Each PROCEED_CONSERVATIVE case must carry its flag forward, and a flagged value is treated as unverified.
- **Accountability.** The source describes this component as decision-support inside a human-in-the-loop architecture. Confirm with compliance that unattended operation is acceptable under the lender's own policy and the applicable RBI requirements before going live.

Safeguards that do not block a case: a circuit breaker that, when below_threshold_rate, gate_trigger_counts or the resubmission rate drifts past its limit, stops issuing PROCEED, holds new cases in an automatic retry queue without contacting borrowers, and alerts engineering about the system (not about individual cases); a full provenance and audit trail for every decision; and optional after-the-fact sampling of auto-processed cases, which nothing waits on and whose results become production labels. The agent is fully automated without the sampling.

11. Future Evaluation Scope

- **Multilingual KYC extraction:** benchmark performance across Hindi, Tamil, Telugu, and Marathi regional identity documents.
- **Per-segment slicing:** evaluate performance by employer tier, income band, scan resolution, and bank statement format.
- **Latency and cost:** tokens, wall-clock time, and OCR compute cost per multi-page document bundle, including the cost of recovery passes.
- **Resubmission simulation:** replay borrower replies so the benchmark can score final outcomes as well as first-pass outcomes.

12. Implementation, Reproducibility & Governance Note

The existing commands still reproduce the baseline:

```
python -m src.evaluation.eval_cli
python -m src.evaluation.eval_cli --output evaluation_report.json
python -m pytest tests/ -v
```

The evaluation CLI described in the source reports the retired HITL metrics. The new metrics in section 7 need to be added to it, and the expected-outcome labels from section 2 added to the dataset. The agent's gate logic also has to change to implement the automated actions; this report specifies the metrics and does not change the code.

The Document Ingestion Agent remains an evidentiary extraction and reconciliation component, but in this version it acts without a reviewer. The ±5% tolerance, the salary variance limit, the cost matrix and every target marked as proposed are engineering choices; they should not be represented as official RBI rules or regulatory requirements.